Semester	B.E. Semester VII – CMPN
Subject	R-Programming
Subject Professor In-charge	Prof. Sanjeev Dwivedi
Assisting Teachers	Prof. Sanjeev Dwivedi
Laboratory	

Student Name	Sakshi Patil	
Roll Number	23102A0046	
Grade and Subject Teacher's Signature		

Experiment Number	07
Experiment Title	Customer Segmentation and Predictive Analytics Using Machine Learning (R)
Objectives (Skill Set / Knowledge Tested / Imparted)	• To clean and pre-process real-world e-commerce transaction data (missing customer IDs, cancellations, duplicates, outliers) • To engineer customer-level RFM features (Recency, Frequency, Monetary) and additional behavioural features • To segment customers using K-Means (Elbow Method) and compare it with Hierarchical (Ward) clustering using the Silhouette Score • To visualise customer segments using PCA (2D) and 3D plots, and to profile each cluster to identify the high-value segment • To train Random Forest and SVM models to predict high-value customers and compare them using Accuracy, Precision, Recall, F1-score, ROC-AUC and Confusion Matrix • To analyse feature importance and to propose segment-wise targeted marketing recommendations
Description	This experiment implements an end-to-end customer analytics pipeline in R for an online retailer. It uses the UCI Online Retail dataset (541,909 transactions of a UK-based e-commerce company, 01/12/2010 – 09/12/2011) with the columns InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. The data was first cleaned by removing rows with missing CustomerID, cancelled invoices (InvoiceNo starting with "C"), non-positive quantity / price and duplicate rows, leaving 392,692 transactions of 4,338 customers. Six customer-level features were then engineered: Recency, Frequency, Monetary, AverageTransactionValue, TotalQuantity and PurchaseFrequency. Because the features are highly right-skewed, they were winsorised at the 1st and 99th percentile, log(1 + x) transformed and standardised before clustering. The number of clusters was chosen with the Elbow Method and the average Silhouette Score (k = 3). K-Means and Ward hierarchical clustering were compared using the Silhouette Score and the Adjusted Rand Index, and the segments were

	visualised with PCA and scatter plots. Each cluster was profiled using the original (un-transformed) values and named (Champions, Loyal Customers, At-Risk / Lost). Finally, a binary target “HighValue” was created from the high-value cluster, and Random Forest and SVM (RBF kernel) classifiers were trained on a stratified 75/25 split and compared. Feature importance (Gini and permutation) was analysed and targeted marketing recommendations were prepared for every segment.
Program	**GitHub link:** <paste your GitHub repository link here> Language / Platform: R (Google Colab R runtime) Packages: readxl, dplyr, tidyr, ggplot2, cluster, randomForest, e1071, pROC, plotly, htmlwidgets
Output	**1. Data pre-processing and feature engineering**

Step	Rows remaining
Original rows	541,909
After removing missing CustomerID	406,829
Cancelled invoices removed	8,905
After removing invalid quantity / price	397,884
After removing duplicates	392,692
Unique customers	4,338

Customer-level table: 4,338 customers × 6 features. Mean Recency = 93.06 days, mean Frequency = 4.27 invoices, mean Monetary = £2,048.69 (median £668.57), showing strong right skew.

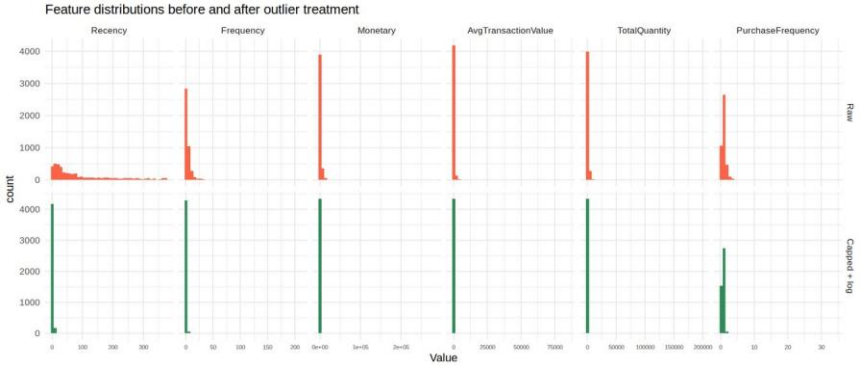


Figure 1: Feature distributions before (raw) and after outlier treatment (winsorise + log)

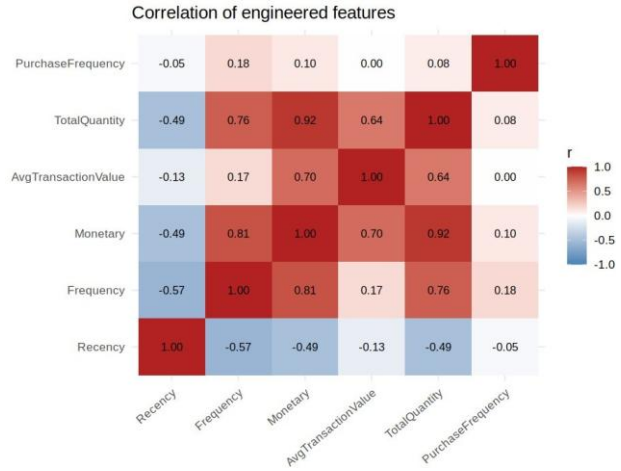


Figure 2: Correlation of engineered features

2. Optimal number of clusters (Elbow Method and Silhouette)

k	2	3	4	5	6	7	8	9	10
WSS	16368.7	13198.4	11248.2	10005.3	8947.9	8139.4	7602.2	7130.7	6700.8
Silhouette	0.3265	0.2649	0.2517	0.2407	0.2578	0.2500	0.2453	0.2346	0.2330

Selected number of clusters: k = 3 (best Silhouette among k = 3 to 6).

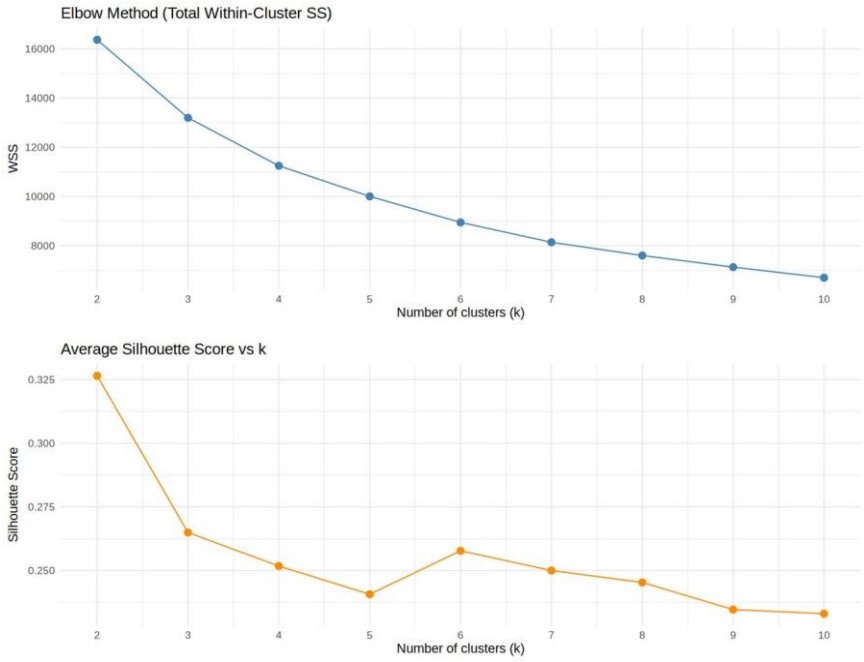


Figure 3: Elbow curve (left) and average Silhouette Score vs k (right)

3. K-Means vs Hierarchical clustering

Method	Clusters	Silhouette Score
K-Means	3	0.2649
Hierarchical (Ward)	3	0.2042

K-Means cluster sizes: 1754, 1816, 768. Hierarchical cluster sizes: 1418, 1237, 1683.
Adjusted Rand Index between the two segmentations = 0.3929. K-Means has the higher Silhouette Score, so it was used for profiling and prediction.

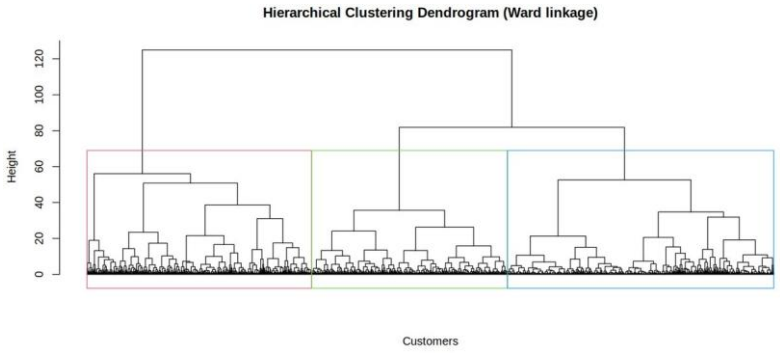


Figure 4: Hierarchical clustering dendrogram (Ward linkage, k = 3)

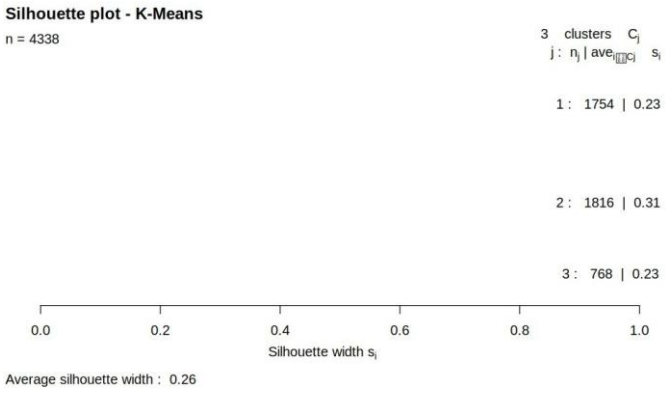
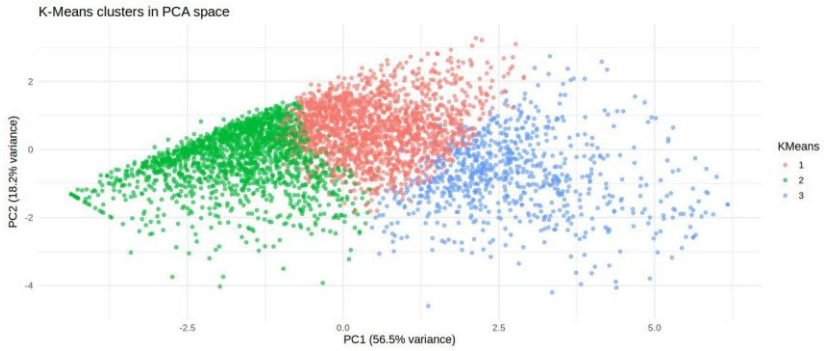


Figure 5: Silhouette plot – K-Means

4. PCA visualisation

Explained variance: PC1 = 56.5%, PC2 = 18.2%, total = 74.6%. Monetary (0.530), TotalQuantity (0.516) and Frequency (0.457) drive PC1.



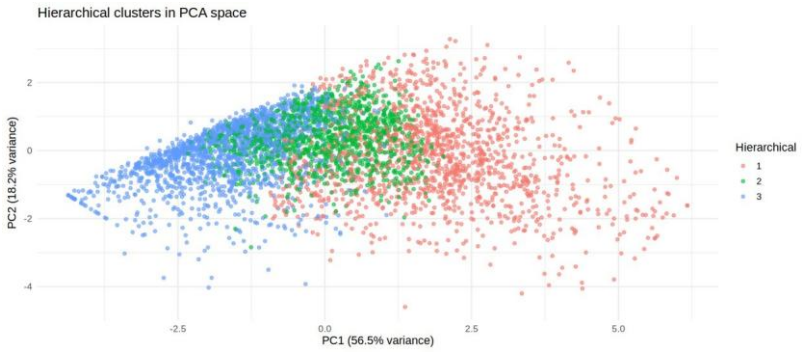


Figure 6: K-Means clusters (top) and Hierarchical clusters (bottom) in PCA space

5. Cluster profiling (original values)

Segment	Cust.	% Rev.	Recency	Freq.	Monetary	Avg Txn	Qty
C1: Loyal Customers	1754	14.0	63.36	3.31	1325.69	515.74	819.16
C2: At-Risk / Lost	1816	3.0	152.20	1.49	287.41	213.26	163.04
C3: Champions (High-Value)	768	83.0	21.06	13.06	7864.60	676.89	4451.97

Cluster 3 (Champions) is the high-value segment: only 17.7% of customers but 83% of total revenue.

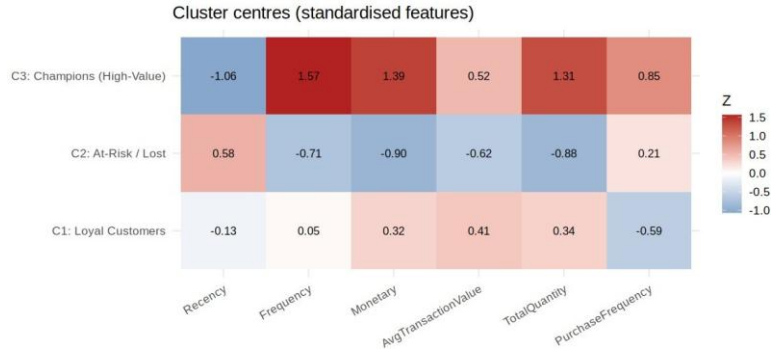
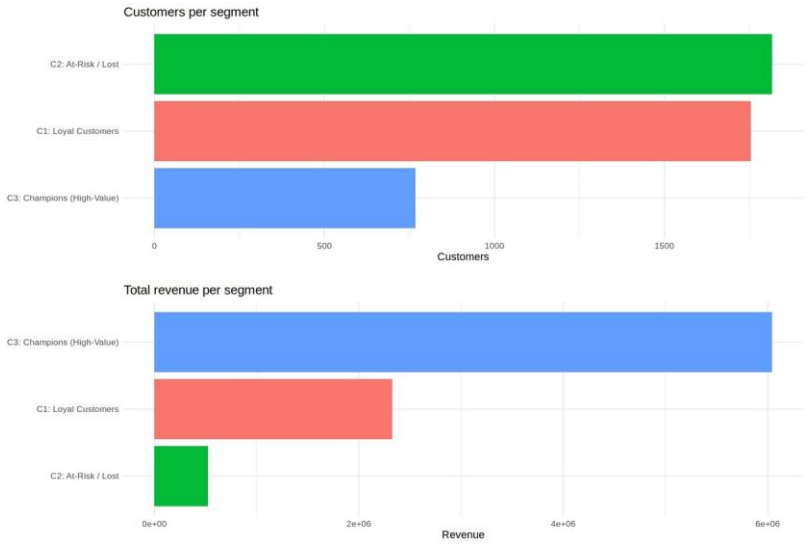


Figure 7: Cluster centres (standardised features)



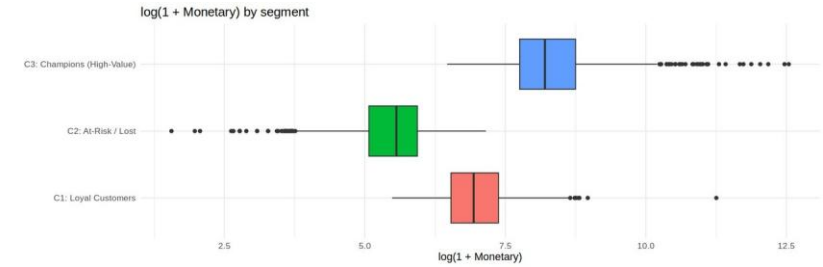


Figure 8: Customers, total revenue and log(1 + Monetary) by segment

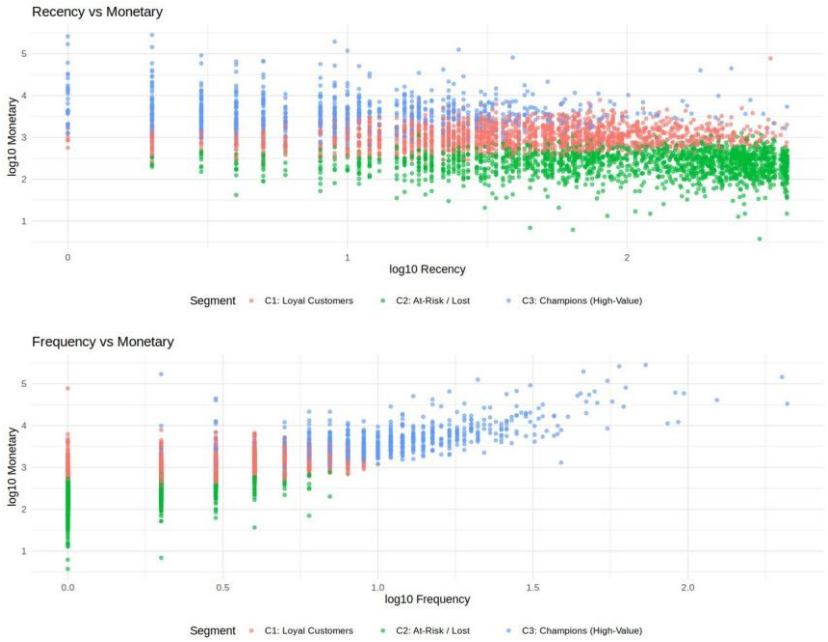


Figure 9: Recency vs Monetary and Frequency vs Monetary (static views of the 3D plot)

6. Predicting high-value customers (stratified 75/25 split: Train 3254, Test 1084)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.9779	0.9615	0.9115	0.9358	0.9982
SVM (RBF)	0.9926	0.9792	0.9792	0.9792	0.9998

Confusion matrices (rows = actual, columns = predicted): Random Forest – [885, 7; 17, 175] SVM – [888, 4; 4, 188].
Best model based on F1-Score: SVM (RBF).

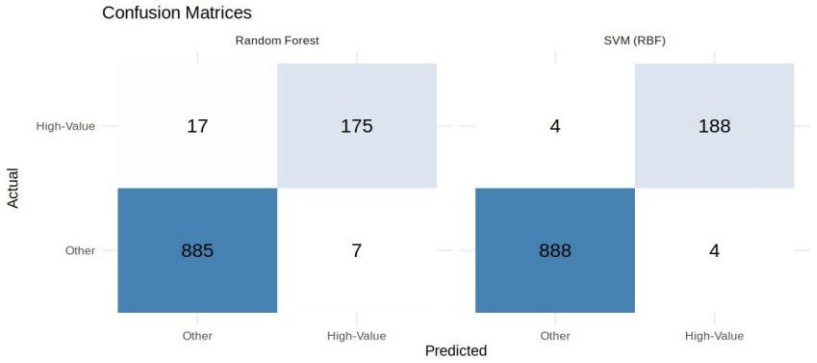


Figure 10: Confusion matrices of Random Forest and SVM

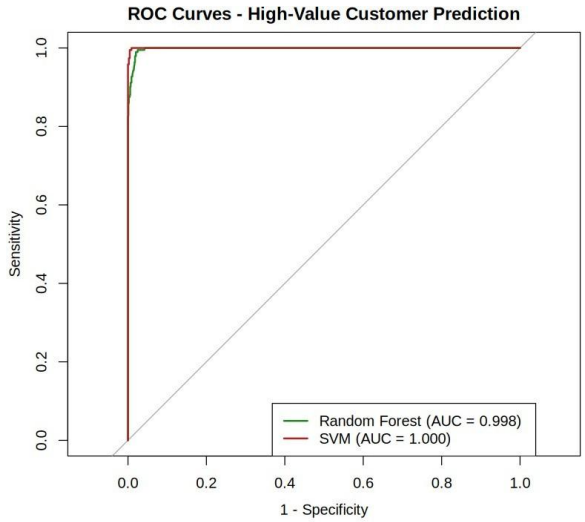


Figure 11: ROC curves (Random Forest AUC = 0.998, SVM AUC = 0.9998)

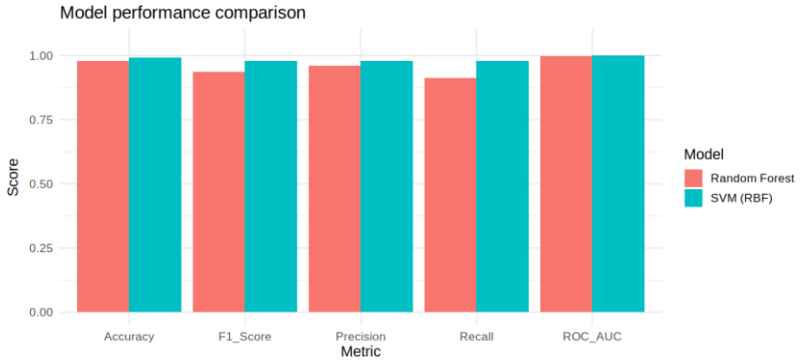
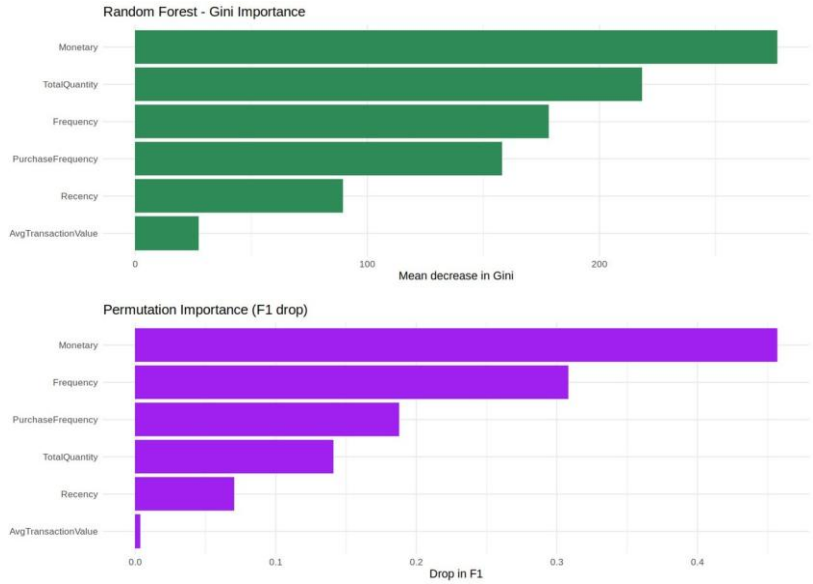


Figure 12: Model performance comparison

7. Feature importance (Random Forest)

Feature	Gini	Permutation (F1 drop)
Monetary	276.56	0.4567
TotalQuantity	218.36	0.1409
Frequency	178.20	0.3081
PurchaseFrequency	157.93	0.1877
Recency	89.37	0.0704
AvgTransactionValue	27.28	0.0036

	Monetary is the most important feature for identifying high-value customers.  *Figure 13: Gini importance and permutation importance* **8. Segment-wise marketing recommendations** • **Champions (High-Value):** VIP / loyalty programme, early access and personalised bundles; ask for reviews and referrals; avoid heavy discounting. • **Loyal Customers:** cross-sell and up-sell using purchase history; reward repeat purchases with points and tiers to move them towards Champions. • **At-Risk / Lost:** time-limited win-back campaigns, surveys to learn why they left, and capped marketing spend (stop after 2–3 attempts with no response).
Conclusion	The R program successfully implemented an end-to-end customer segmentation and predictive analytics pipeline on the UCI Online Retail dataset. After cleaning 541,909 transactions down to 392,692 valid records of 4,338 customers, six RFM-based features were engineered, outlier-treated, log-transformed and standardised. The Elbow Method and Silhouette Score gave k = 3, and K-Means (Silhouette 0.2649) outperformed Hierarchical Ward clustering (Silhouette 0.2042), so K-Means segments were used. Profiling showed that the Champions (High-Value) segment, only 17.7% of customers (768), contributes 83% of the total revenue, with the lowest Recency (21 days) and highest Frequency (13.06) and Monetary value (£7,864.60). Loyal Customers form a middle group (40.4% of customers, 14% of revenue), while the At-Risk / Lost group (41.9% of customers) contributes only 3% of revenue. Both Random Forest and SVM predicted high-value customers very accurately; SVM was the best model (Accuracy 0.9926, F1 0.9792, ROC-AUC 0.9998) compared to Random Forest (Accuracy 0.9779, F1 0.9358, ROC-AUC 0.9982). Feature importance showed that Monetary, TotalQuantity and Frequency are the strongest indicators. Since the target is derived from clusters built on the same features, high scores are expected. The experiment gave practical skills in RFM analysis, clustering, PCA, classification and model evaluation, and the segment-wise recommendations help retain Champions, grow Loyal customers and target

	marketing spend efficiently.